AP Statistics **Unit 1, Lesson 1.3**

Group Activity: Frequency Tables & Fair Comparisons

Name: ____________________________ Partner: _______________________ Period: _________

Part 1 (group, 8 min): Use the raw data below to build a frequency and relative frequency table. Then write a description of the distribution.

Part 2 (group, 5 min): A second dataset of different size is provided. Compare the two distributions using relative frequency.

Part 3 (individual, 3 min): Reflection question.

Part 1 Build the Table

A survey of **30 middle school students** asks: "What is your favorite sport?" Results are listed below.

Soccer	Basketball	Baseball	Soccer	Swimming	Tennis
Baseball	Soccer	Swimming	Basketball	Soccer	Soccer
Tennis	Basketball	Soccer	Swimming	Baseball	Basketball
Soccer	Tennis	Basketball	Soccer	Swimming	Baseball
Basketball	Soccer	Tennis	Basketball	Soccer	Baseball

Sport	Tally	Frequency	Rel. Frequency
Baseball			
Basketball			
Soccer			
Swimming			
Tennis			
Total		____________	_______________

1. Does your total equal 30? ______ Does your relative frequency column sum to 1? ______
2. What is the most popular sport and what percentage of students chose it?
3. Write a complete sentence describing the distribution of favorite sport for these 30 students. Include the most and least common categories.

Part 2 Compare Two Groups

A separate survey of **50 high school students** asks the same question. The frequency table is partially completed. Fill in the relative frequency column and then answer the comparison questions.

Sport	Frequency	Rel. Frequency
Baseball	8	__________
Basketball	15	__________
Soccer	14	__________
Swimming	7	__________
Tennis	6	__________
Total	50	__________

1. A student says: "Soccer is the most popular sport in both groups because both tables show soccer near the top." Is this a valid comparison? Why or why not?

2. **Using relative frequencies**, which group — middle school or high school — shows a stronger preference for basketball? Show your evidence.

3. Suppose a third school has 200 students and 60 choose soccer. Is soccer more popular there than in the middle school group? Explain using relative frequency.

Part 3 Individual Reflection

In your own words: Why is a relative frequency table more useful than a frequency table when comparing two groups? Give a specific example from today's activity to support your answer.